

Dat Mai

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EDUCATION

Georgia Institute of Technology

GPA: 3.65

Bachelor of Science in Computer Science–Intelligence, InfoNetworks

Graduation: December 2026

Relevant Coursework: Machine Learning, Artificial Intelligence, Computer Systems and Networks, Database Structures, Enterprise Computing, Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, HTML/CSS, JavaScript, TypeScript

Technologies: PyTorch, pandas, scikit-learn, OpenCV, Jupyter Notebooks, HuggingFace, LangChain, CUDA, Matplotlib, NumPy, FastAPI, React, Git, Docker, Jira

EXPERIENCE

Georgia Tech Robotic Musicianship Lab

Jan 2025 – May 2026

Machine Learning Undergraduate Researcher

Atlanta, GA

- Developed real-time multimodal generative audio system (text & accompaniment → rap vocal output) based on state-of-the-art models in PyTorch, enabling synchronous human–robot rap interactions
- Engineered large-scale preprocessing pipeline for 90,000+ songs producing a structured RapBank dataset optimized for low-latency training and inference
- Implemented low-latency inference engine for zero-shot vocal generation and real-time audio mixing, serving results via FastAPI/Uvicorn to a React frontend with real-time factor ≈ 1.0

DoReMi Music School

May 2025 – Aug 2025

Software Developer and Technical Consultant

Lilburn, GA

- Redesigned registration, scheduling, and billing workflows via stakeholder interviews; built Python/Tkinter/SQLite desktop app, reducing Excel errors by $\approx 78\%$
- Automated monthly invoicing, cutting processing time from **3–4 hours** to **10 minutes**, enabling higher weekly student capacity and directly improving profits
- Implemented Google-Drive for document storage and automated SMS reminders, prioritizing ease of use for a non-technical client while cutting manual follow-ups by **70%** and improving on-time payment rates

PROJECTS

ShinyHunterBot

Feb 2026 – Mar 2026

- Built end-to-end CV pipeline for automated shiny Pokemon detection in real-time emulator gameplay using Python, Pytorch, OpenCV
- Designed custom CNN with frame aggregation across 60-frame encounter sequences, achieving 100% classification accuracy and < 0.02 BCE validation loss
- Engineered automated data collection pipeline with configurable batch control, producing cleanly labeled and structured encounter datasets ready for training with zero manual intervention

Diabetic Retinopathy Detector

Sept 2025 – Dec 2025

- Developed 5-class diabetic retinopathy severity classifier via transfer learning with ResNET50 on the highly imbalanced APTOS 2019 dataset, achieving kappa > 0.88
- Implemented strategic layer freezing and aggressive data augmentation to prevent overfitting and maximize transfer learning effectiveness on limited medical data (3600+ images)
- Engineered clean, production-grade training infrastructure with PyTorch Lightning, including class-weighted CE loss to counter class imbalances, ReduceLR scheduling, and custom epoch-level kappa tracking

WanderSync: Collaborative Travel

Aug 2024 – Dec 2024

- Co-led Android app development (Java/XML, MVVM) for 6-person team using Firebase for auth/storage, Git for version control, and Trello for Agile/Scrum sprints, applying SOLID/GRASP principles
- Enabled user account creation, vacation planning, restaurant/accommodation reservations, and multi-user travel coordination, enhancing collaborative trip management